

1 First Refinements Computer Lab

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Graph 1

```
require(Lock5Data)
```

```
## Loading required package: Lock5Data
```

```
data(MarriageAges)
```

```
head(MarriageAges)
```

```
##   Husband Wife
## 1      53    50
## 2      38    34
## 3      46    44
## 4      30    36
## 5      31    23
## 6      26    31
```

```
gf_point(Husband~Wife,data=MarriageAges, alpha = 0.7)%>%gf_labs(x = "Wife Age", y = "Husband Age", title = "Marriage Ages")
```

```
## Warning: All aesthetics have length 1, but the data has 105 rows.
```

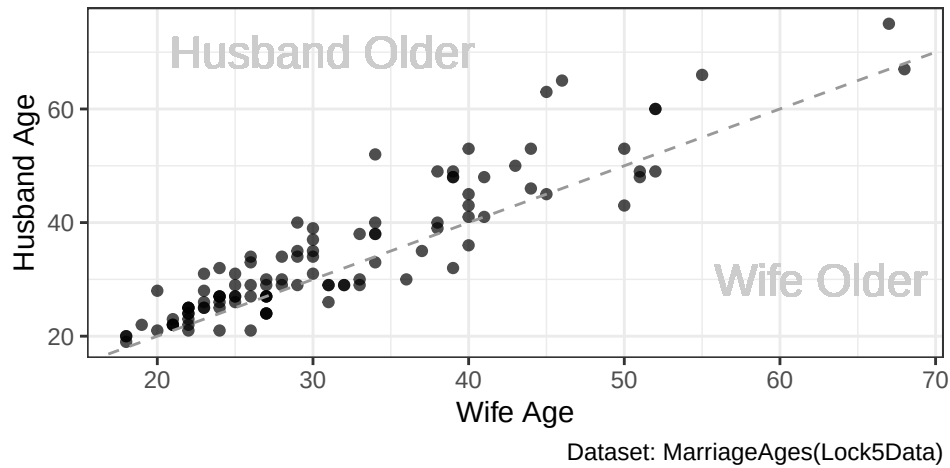
```
## i Please consider using `annotate()` or provide this layer with data containing
##   a single row.
```

```
## All aesthetics have length 1, but the data has 105 rows.
```

```
## i Please consider using `annotate()` or provide this layer with data containing
##   a single row.
```

Husbands Tend to Marry Younger Slightly

Examining Relationship between Husband Age explained by Wife



Graph 2

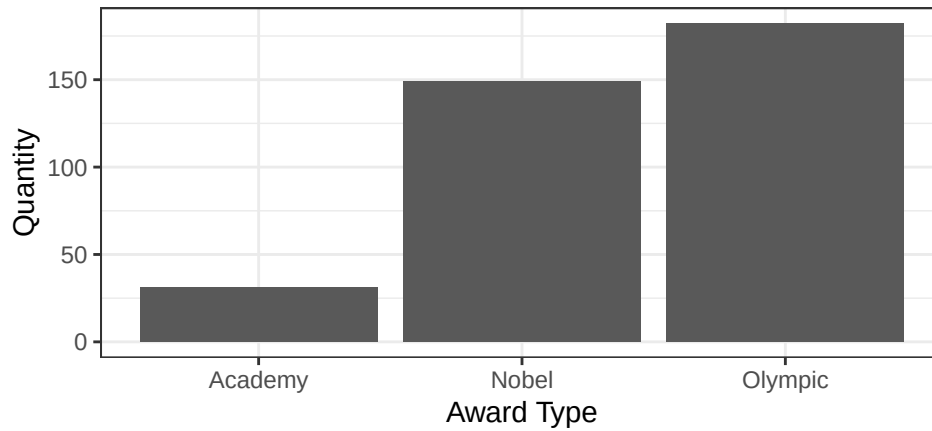
```
data(StudentSurvey)
head(StudentSurvey)
```

```
##      Year Sex Smoke  Award HigherSAT Exercise TV Height Weight Siblings
## 1   Senior  M   No Olympic      Math      10  1   71   180         4
## 2 Sophomore F   Yes Academy      Math       4  7   66   120         2
## 3 FirstYear M   No  Nobel      Math      14  5   72   208         2
## 4   Junior  M   No  Nobel      Math       3  1   63   110         1
## 5 Sophomore F   No  Nobel    Verbal       3  3   65   150         1
## 6 Sophomore F   No  Nobel    Verbal       5  4   65   114         2
## BirthOrder VerbalSAT MathSAT SAT GPA Pulse Piercings
## 1          4       540     670 1210 3.13   54        0
## 2          2       520     630 1150 2.50   66        3
## 3          1       550     560 1110 2.55  130        0
## 4          1       490     630 1120 3.10   78        0
## 5          1       720     450 1170 2.70   40        6
## 6          2       600     550 1150 3.20   80        4
```

```
gf_bar(~ Award , data = StudentSurvey )>%gf_labs(x = "Award Type", y = "Quantity", title="Olympic Award")
```

Olympic Award is the preferred award among students

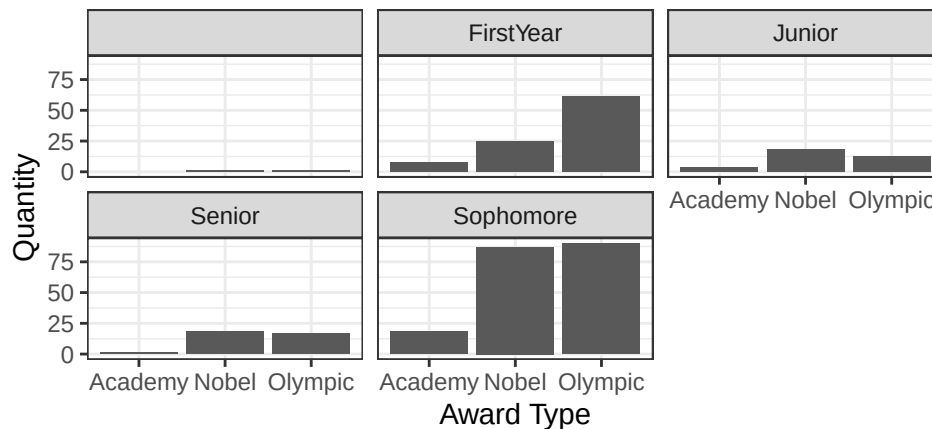
Examining Student Preferences



```
gf_bar(~ Award , data = StudentSurvey )>% gf_facet_wrap( ~ Year )>% gf_labs(x = "Award Type", y = "Quantity")
```

Olympic Award tends to be the preferred award

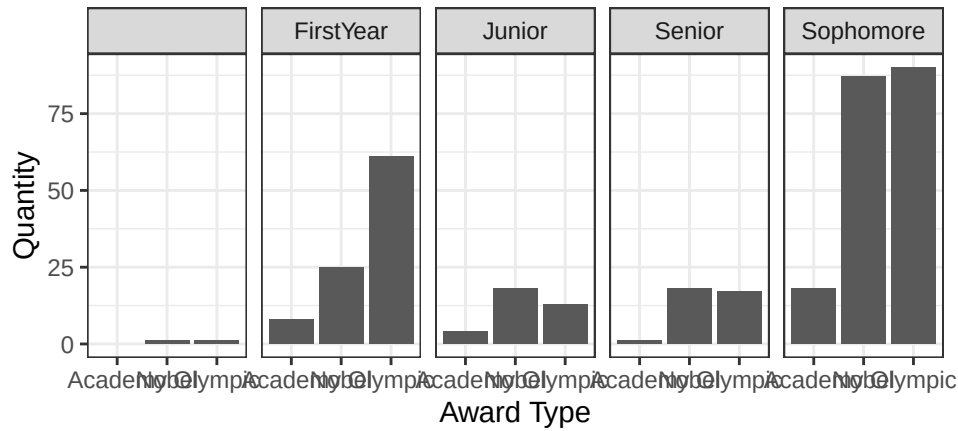
Examining Student Preferences – Facet



```
gf_bar(~ Award , data = StudentSurvey )>% gf_facet_wrap( ~ Year, nrow=1 )>% gf_labs(x = "Award Type", y = "Quantity")
```

Olympic Award tends to be the preferred award

Examining Student Preferences – Facet nrow=1

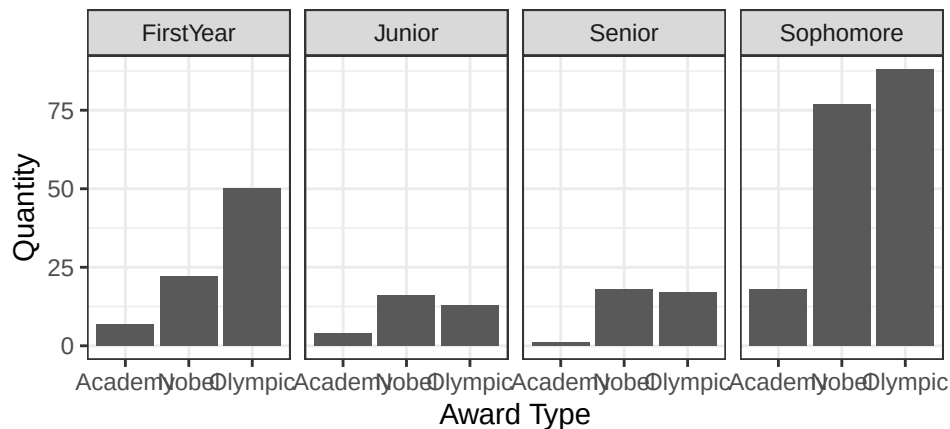


Dataset: StudentSurvey(Lock5Data)

```
gf_bar(~ Award , data = na.omit(StudentSurvey) )>% gf_facet_wrap( ~ Year,nrow=1 )>% gf_labs(x = "Award
```

Olympic Award tends to be the preferred award

Examining Student Preferences



Dataset: na.omit(StudentSurvey)(Lock5Data)

Graph 3

```
require(resampledData)
```

```
## Loading required package: resampledData
```

```
##
```

```
## Attaching package: 'resampledData'
```

```
## The following object is masked from 'package:datasets':
```

```
##
```

```
## Titanic
```

```
data(Cuckoos)
```

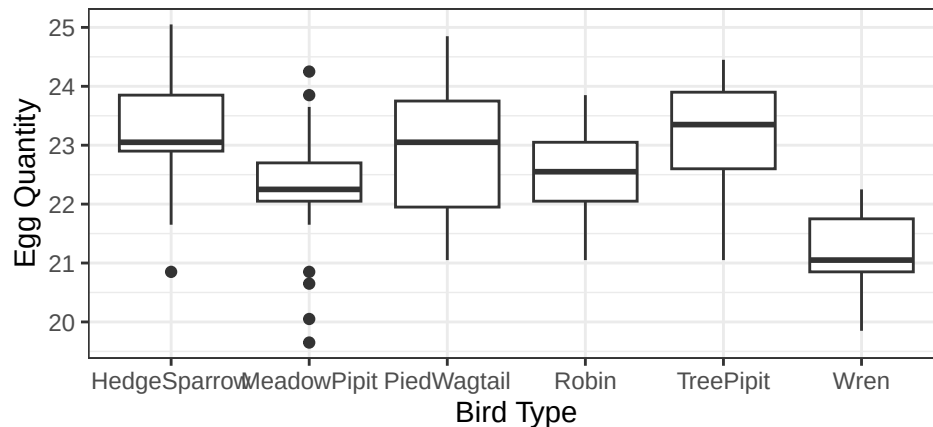
```
head(Cuckoos)
```

```
## ID Eggs Bird
## 1 1 19.65 MeadowPipit
## 2 2 20.05 MeadowPipit
## 3 3 20.65 MeadowPipit
## 4 4 20.85 MeadowPipit
## 5 5 21.65 MeadowPipit
## 6 6 21.65 MeadowPipit
```

```
gf_boxplot(Eggs ~ Bird , data = Cuckoos )>%gf_labs(x = "Bird Type", y = "Egg Quantity", title="Showcas")
```

Showcasing Birds and Their Egg Quantity

* Notable outliers for Meadow Pipits and Hedge Sparrow



Dataset: Cuckoos(resampleddata)

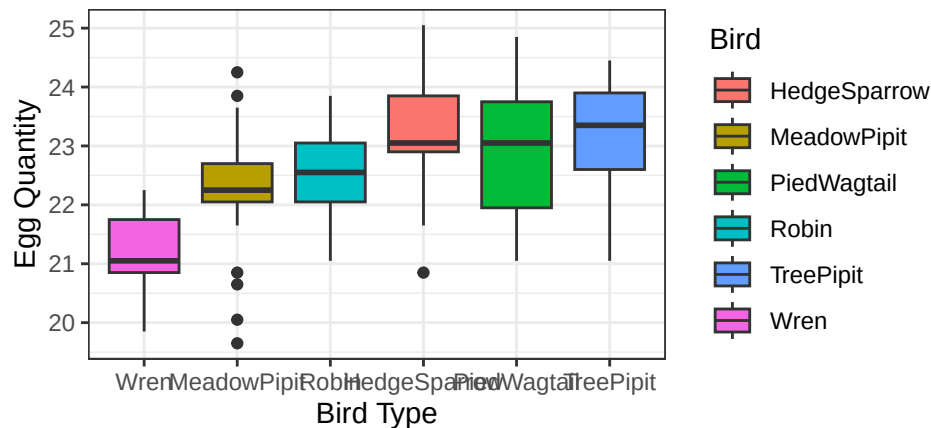
```
require( forcats )
```

```
## Loading required package: forcats
```

```
gf_boxplot( Eggs ~ fct_reorder( Bird , Eggs ) , data = Cuckoos, fill = ~ Bird)>%gf_labs(x = "Bird Type")
```

Showcasing Birds and Their Egg Quantity

* Notable outliers for Meadow Pipits and Hedge Sparrow

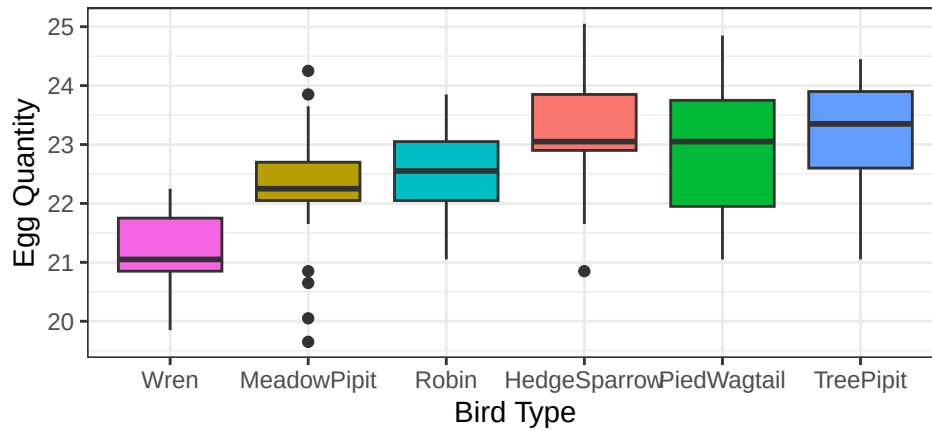


Dataset: Cuckoos(resampleddata)

```
gf_boxplot( Eggs ~ fct_reorder( Bird , Eggs ) , data = Cuckoos, fill = ~ Bird , show.legend = FALSE)>%
```

Showcasing Birds and Their Egg Quantity

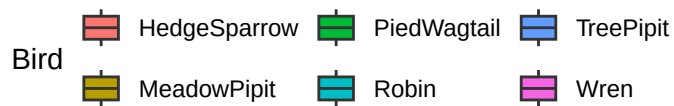
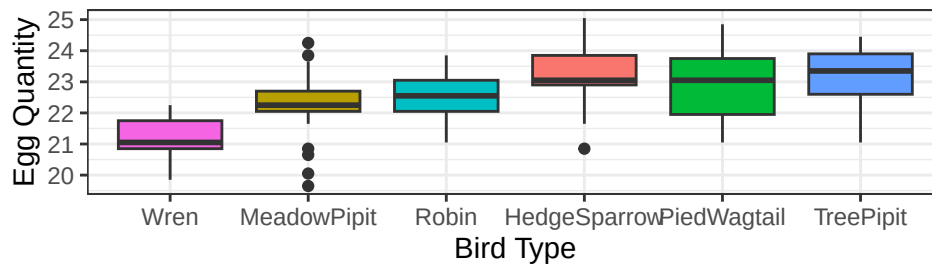
* Notable outliers for Meadow Pipits and Hedge Sparrow



```
gf_boxplot( Eggs ~ fct_reorder( Bird , Eggs ) , data = Cuckoos, fill = ~ Bird)%>%gf_theme(legend.position="bottom")
```

Showcasing Birds and Their Egg Quantity

* Notable outliers for Meadow Pipits and Hedge Sparrow



Dataset: Cuckoos(resampleddata)